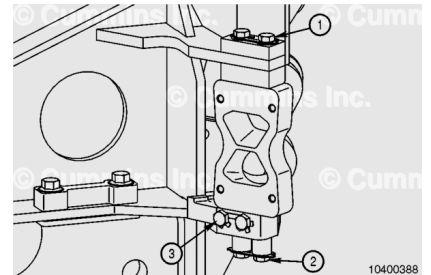


Trainings ▾

General Information

The capscrews shown will be referred throughout this procedure as:

1. Outboard Horizontal.
2. Inboard Horizontal.
3. Inboard Vertical.



Preparatory Steps

Rail Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not rescued, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

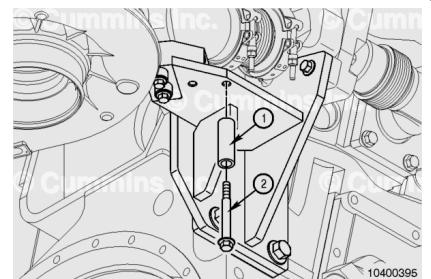
⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

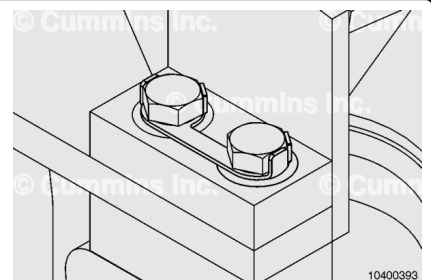
- Remove the intake and exhaust pipes from the turbocharger.
- Remove the turbocharger. Refer to Procedure 010-033 (Turbocharger) in Section 10 (</qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html>)

Remove

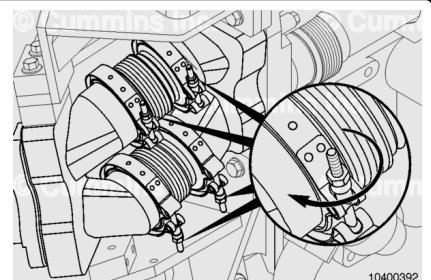
Remove the turbocharger hose support spacer mounting. Disconnect both the existing tube clip support and the tube brace from the single spacer (1).



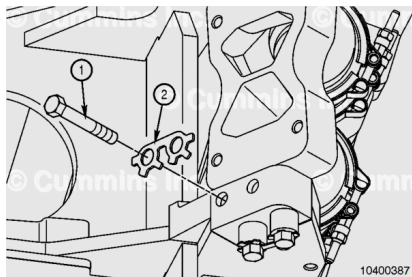
Bend the lock plate tabs down with a chisel to allow capscrew removal.



Remove the bellows V-band clamps.

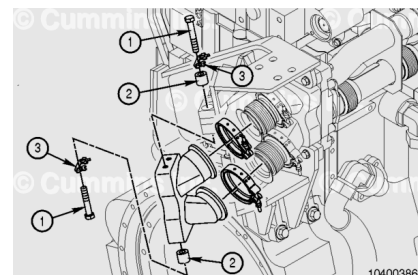


Remove the inboard vertical capscrews (1), and lockplates (2).
Discard the vertical capscrews (1).



Remove the horizontal capscrews (1), spacers (2), and lockplates (3).

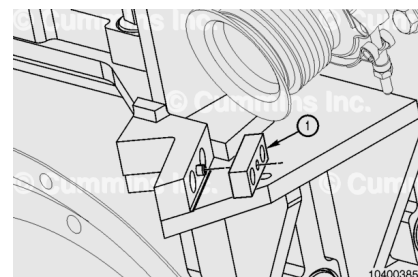
Remove the turbocharger inlet exhaust connection.



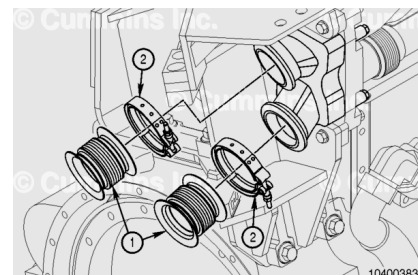
Remove the clamp plate (1).

Remove the roll pin.

Discard the clamping plate (1).

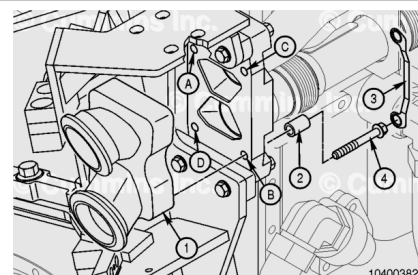


Remove the bellows clamp (2). Remove the bellows (1) from the exhaust transfer connection.

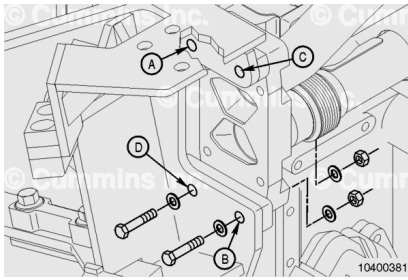


Remove the exhaust transfer connection casting mounting capscrews (4) 32 mm [1.3 in] spacers (2), lockplates (3), and gasket from the exhaust manifold.

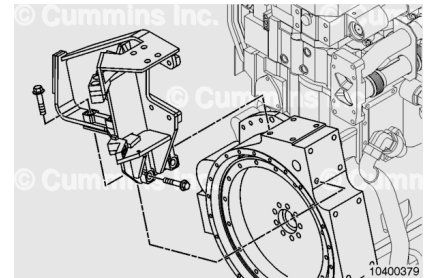
Remove the exhaust transfer casting (1)



Remove the four exhaust manifold to turbocharger mounting bracket nuts, washers, and capscrews.



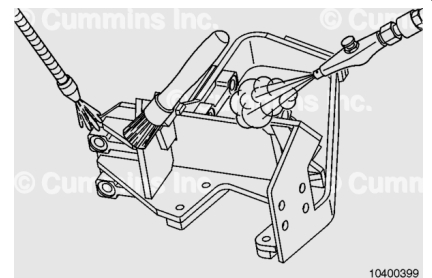
Remove the turbocharger mounting bracket mounting cap screws from the flywheel housing.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

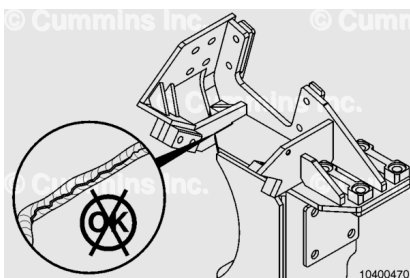
Use a soda blast process to remove paint from the bracket. Do **not** sand blast or bead blast parts, as this will cause damage.

Once the paint is removed, clean the bracket with solvent.

Dry with compressed air.

Use a magnetic particle inspection to inspect the bracket for cracks or other damage.

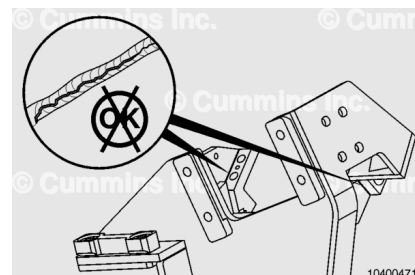
Inspection **must** be performed by an operator holding at least a level 2 ASNT or PCN certification.



Inspect the welds in the highlighted areas closely.

Replace the turbocharger support bracket if any cracks or other damage is found.

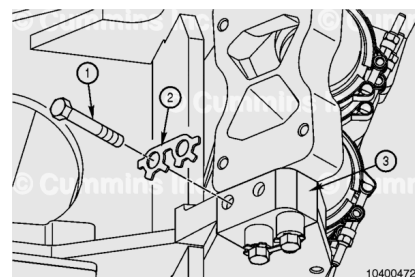
Note : Do not weld or repair the turbocharger support bracket, as this will cause further damage to the bracket.



Inspect the connection brace (3) for cracks and wear damage. Replace if damage is found.

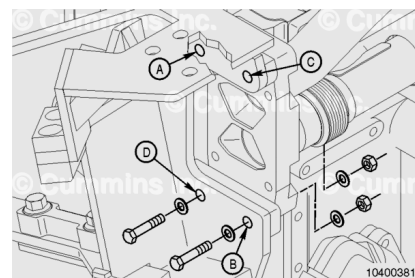
Note : The connection brace (3) must be replaced at engine overhaul.

Note : The vertical capscrews (1) must be replaced anytime the capscrews are removed from the engine.



Inspect the capscrews for cracks and wear damage. Replace if damage is found.

Note : All capscrews must be replaced at engine overhaul.

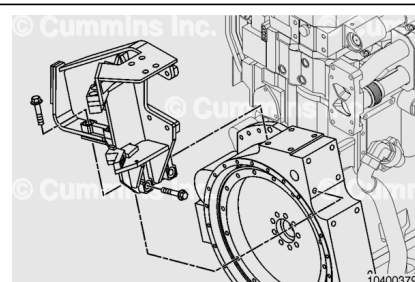


Install

Note : Each time the turbocharger support bracket is installed, the following parts must be replaced:

- Clamping plate
- Two inboard vertical capscrews.

Note : At engine overhaul the following parts must be replaced:



- All fasteners, washers, and lockplates
- Connection brace
- Clamping plate.

Install and tighten the side capscrews first, and then install and tighten the top capscrews.

Torque Value:

Initial Torque

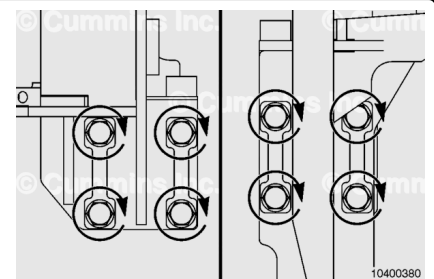
1. 14 n•m [124 in-lb]

Tighten the side capscrews to the final torque value first, then tighten the top capscrews to the final torque value.

Torque Value:

Final Torque

1. 237 n•m [175 ft-lb]



Use a clean cloth and remove any manufacturing oil or preservative on the mating surfaces. Install the exhaust manifold to the turbocharger mounting bracket. Paint is **not** allowed on the mating surfaces. Tighten the capscrews in the sequence shown.

Torque Value:

Initial Torque

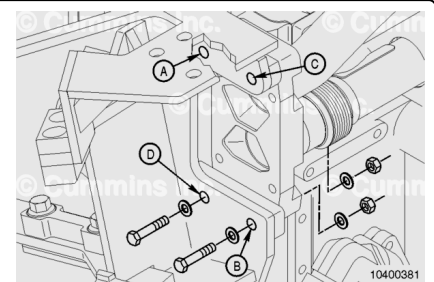
1. 14 n•m [124 in-lb]

Check the mating surfaces for gaps with a 10 mm [0.004 in] feeler gauge. If the feeler gauge can be inserted between the mating surfaces, the turbocharger bracket is damaged. Replace the damaged part and check the flatness. Check the mating surfaces again. If the feeler gauge can **not** be inserted between the mating surfaces, tighten the capscrews in the sequence shown to the final torque.

Torque Value:

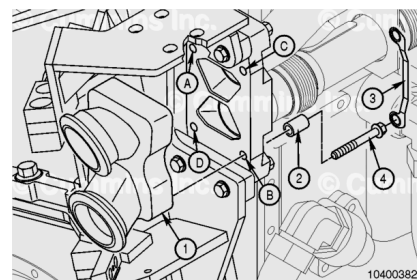
Final Torque

1. 72 n•m [53 ft-lb]



Use a clean cloth and remove any manufacturing oil or preservative on mating surfaces. Paint is **not** allowed on mating surfaces.

Install the gasket and exhaust transfer connection casting (1), to the exhaust manifold flange. Apply Loctite™ 51606 heavy duty anti-seize compound (do **not** substitute) to the first 4 to 5 threads of the capscrew, and under the bearing area of the capscrew head. Do **not** apply an excessive amount of Loctite™ 51606. Use the 33 mm [1.3 in] spacers (2), on capscrews (4) and install. Hand start the capscrews.



Tighten the capscrews in the following sequence. If an angle gage is **not** available, or space restrictions prohibit the use of an angle gauge, use the alternate torque after the initial torque.

Torque Value:

1. 30 n•m [22 ft-lb]
2. Loosen
3. 6.8 n•m [60 in-lb]
4. Advance 60 degrees

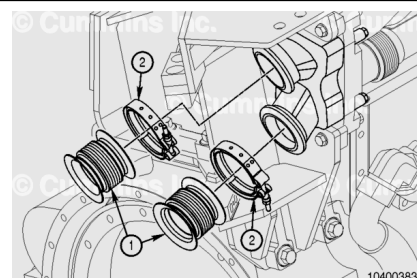
Torque Value:

Alternate Torque

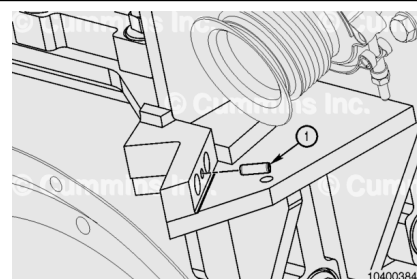
1. 57 n•m [42 ft-lb]

Install the lockplates after tightening.

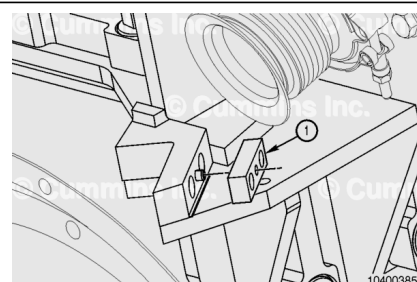
Loosely install the two bellows (1) on the exhaust transfer connection, use V-band clamps (2). The bellows are directional. Install the bellows with the arrow pointing toward the turbocharger. Tighten the V-band clamp **only** enough to prevent the bellows from falling. The bellows **must** be free to move on the casting to align on final assembly.



Install the roll pin (1) that positions the clamp plate for the vertical capscrews on the inboard side lower mount. Gently press the roll pin into the hole in the turbocharger bracket as shown. If necessary, use a plastic faced hammer to gently drive in the roll pin. When finished, the pin will protrude from the turbocharger bracket 6.35 mm [1/4 in].



Install the clamp plate (1) over the roll pin as shown. Gently press the clamp plate down and if necessary, use a plastic faced hammer to gently drive the clamp plate over the pin until the plate is flush with the bracket mating surface. Do **not** leave

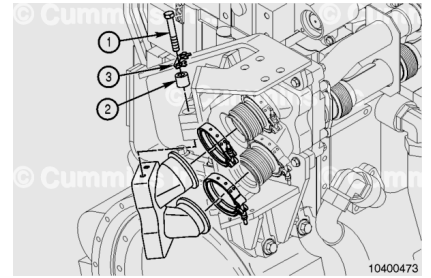


a gap between the plate and the bracket mating surfaces.
Rotate the clamp plate to align the thread holes with the turbocharger bracket bolt holes.

Position the two V-band clamps on the bellows and turbocharger inlet exhaust connection and tighten just enough to position the V-band clamp. Do **not** the tighten the V-band clamp at this time.

Use a clean cloth and remove any manufacturing oil or preservative on mating surfaces. Paint is **not** allowed on mating surfaces.

Install the turbocharger inlet exhaust connection on the turbocharger bracket. Apply Loctite™ 51606 heavy duty anti-seize compound to the first 4 to 5 threads and under the head of the capscrews (1). Install **only** the outboard horizontal capscrews, lockplate (3) and spacers 23 mm [0.9 in] (2). Tighten the capscrews individually in the following sequence.



Torque Value:

1. 92 n•m [68 ft-lb]
2. Loosen and finger tight

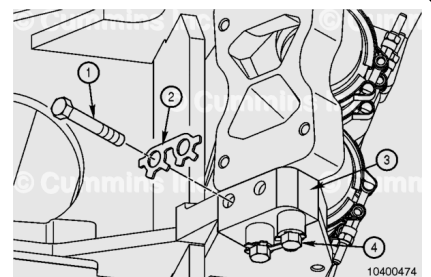
Note : Inboard capscrews are not installed at this time.

Use a clean cloth and remove any manufacturing oil or preservative on mating surfaces. Paint is **not** allowed on the mating surfaces. Apply Loctite™ 51606 heavy duty anti-seize compound to first 4 to 5 threads and under the head of the capscrews.

Install the connection brace (3) to the turbocharger inlet exhaust connection with the inboard horizontal capscrews (4), lockplates (2), and spacers 23 mm [0.9 in]. Tighten finger tight.

Install the connection brace (3) to the turbocharger support bracket with inboard vertical capscrews (1), lockplates (2). Tighten finger tight.

Tighten capscrews individually in the following sequence:



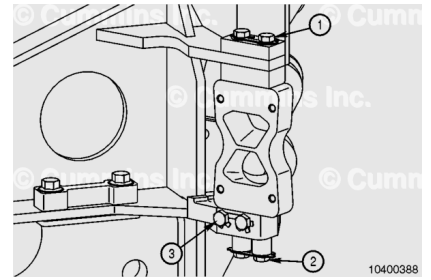
Torque Value:

1. 92 n•m [68 ft-lb]
2. Loosen and finger tight

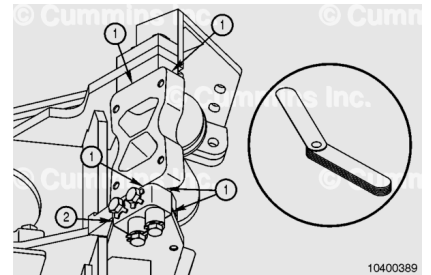
Tighten the capscrews in the sequence shown.

Torque Value:

1. 6.8 n•m [60 in-lb]



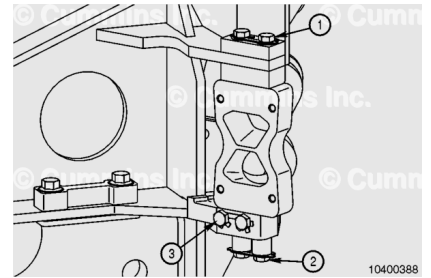
Check mating surfaces (1) for gaps with a 10 mm [0.004 in] feeler gauge. Gaps will reduce the bolted joint preload and can **not** be allowed on mating surfaces. Gaps are allowable on the lower, rear surface of the connection brace and the turbocharger bracket (2), as the capscrew does **not** pass through this surface. This check **must** be performed with the capscrews tightened to 6.8 N•m [60 in-lb].



A step torque process is necessary to maintain even mating of parts surfaces. Tighten the capscrews in the sequence shown.

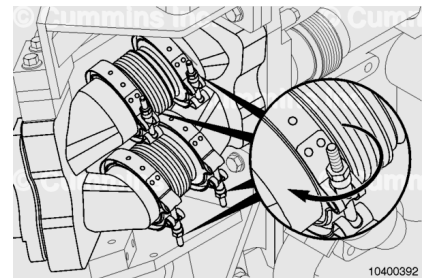
Torque Value: 50 n•m [37 ft-lb]

Torque Value: 92 n•m [68 ft-lb]

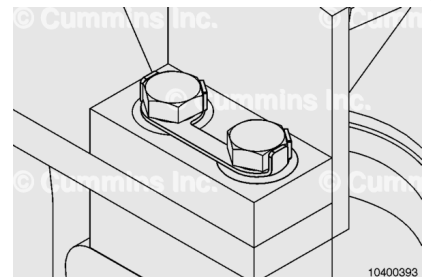


Align the bellows V-band clamps and center the bellows on the cast flange. Make sure the nuts are easy to access and the threaded T-bolt does **not** extend above the top of the turbocharger bracket. Tighten the V-band clamps.

Torque Value: 8.5 n•m [75 in-lb]

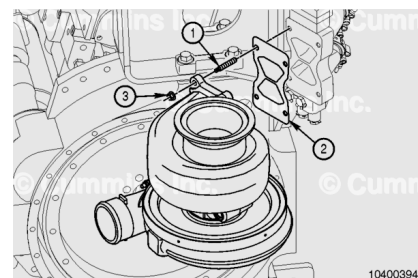


Secure the lockplates on the turbocharger inlet exhaust connection by bending the tabs over the capscrew head flats in two places for each capscrew.



Use a clean cloth and remove any manufacturing oil or preservative on mating surfaces. No paint is allowed on mating surfaces. Apply Loctite™ 51606 heavy duty anti-seize

compound to the first 4 to 5 threads of each end of the 10 mm [0.4 in] stud (1). Thread the stud into the turbocharger inlet exhaust connection with the stud pilot to the outside in all four locations. Position the gasket (2) on the studs and mount the turbocharger. Install the 10 mm [0.4 in] nuts (3) on the studs and tighten finger tight to hold the turbocharger in place.



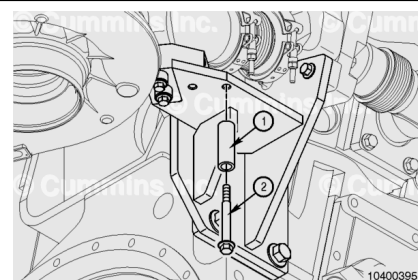
Tighten the nuts diagonally. Check thread protrusion. There should be 2 to 3 threads showing.

Torque Value: 46 n•m [34 ft-lb]

Connect coolant and oil hoses to turbocharger.

Torque Value: 25 n•m [221 ft-lb]

Connect both the existing tube clip support and the tube brace to the spacer (1) and the capscrew (2) to secure. Check the routing of turbocharger oil and coolant hoses: There should be 5 mm [0.2 in] of clearance between hoses and any surface. If **not**, adjust clips and/or bend tube braces to provide clearance.



Torque Value: 46 n•m [34 ft-lb]

Finishing Steps

Rail Applications

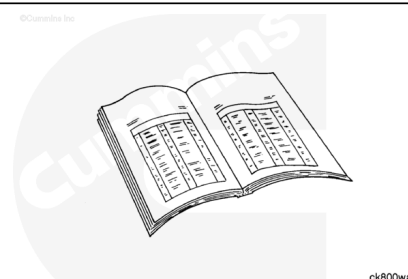
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when filling the cooling system that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

⚠ CAUTION ⚠



Use caution when filling the lubricating system that the oil is not spilled into the bilge area. Do not pump oil overboard. If the oil is spilled, it must be disposed of in accordance with local environmental regulations

- Install the intake and exhaust pipes to the turbocharger.
- Paint the turbocharger bracket assembly with a Cummins Inc. approved paint, such as service tool, Part Number 3824515.
- Operate the engine at 71°C [160°F] coolant temperature and check for leaks.
- Check the V-band clamp torque 8.5 N•m [75 in-lb] after operating the engine.